MMDU MULLANA OFFICIAL SYLLABUS • BMAT-001 • UNIT 2

Unit 2: Unit II: Calculus (10 Hours)

Subject: Mathematics-I (BMAT-001) • **Exam Weightage:** 15 Marks (3×1M + 1×2M + 1×4M + 1×6M) • **Study Time:** 35 mins

Complete Unit II coverage per MMDU BMAT-001 syllabus: Rolle's Theorem, Mean value theorems, Taylor's and Maclaurin's series, Expansion by use of known series, Curvature, Radius of Curvature for Cartesian, Parametric and polar curves, Centre of Curvature, chord of Curvature. Indeterminate forms and L'Hospital's rule, Beta and Gamma Functions and their properties.

1. Mean Value Theorems & Power Series Expansions

1. In-Depth Analysis of the Three Fundamental Mean Value Theorems

Differential calculus is built on three interconnected foundation stones: Rolle's Theorem, Lagrange's Mean Value Theorem, and Cauchy's Generalized Mean Value Theorem.

A. Rolle's Theorem (1691)

✅ **FUNDAMENTAL LAW / RULE:**
Theorem Statement: Let $f : [a, b] \to \mathrm{R}$ be a real function satisfying:

1. f is **continuous** on the closed interval $[a, b]$.

2. f is **differentiable** on the open interval (a, b).

3. $f(a) = f(b)$.

> Then there exists at least one point $c \in (a, b)$ such that:

$$\mathbf{f'(c) = 0}$$

- **Geometric Interpretation:** If a continuous smooth curve starts and ends at the same horizontal height, there must be at least one crest or trough where the tangent line is perfectly horizontal (slope = 0).

- **Physical Interpretation:** If a car returns to its starting point after a road trip, its instantaneous velocity must have been zero at least once when turning around.

B. Lagrange's Mean Value Theorem (First MVT)

✅ **FUNDAMENTAL LAW / RULE:**
Theorem Statement: Let $f : [a, b] \to \mathrm{R}$ be continuous on $[a, b]$ and differentiable on (a, b). Then there exists at least one point $c \in (a, b)$ such that:

$$\mathbf{f'(c) = \frac{f(b) - f(a)}{b - a}}$$

- **Geometric Interpretation:** There exists at least one interior point c where the instantaneous tangent line is strictly parallel to the chord (secant line) joining the endpoints $(a, f(a))$ and $(b, f(b))$.

- **Alternative Form:** Setting $b = a + h$ and $c = a + \theta h$ with $0 < \theta < 1$:

$$f(a + h) = f(a) + hf'(a + \theta h)$$

C. Cauchy's Generalized Mean Value Theorem

✅ **FUNDAMENTAL LAW / RULE:**
Theorem Statement: Let two functions $f, g : [a, b] \to \mathrm{R}$ be continuous on $[a, b]$, differentiable on (a, b), and $g'(x) \neq 0$ for all $x \in (a, b)$. Then there exists at least one $c \in (a, b)$ such that:

$$\mathbf{\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}}$$

- **Proof Architecture:** Define auxiliary function $\phi(x) = f(x) - \lambda g(x)$. Choose $\lambda = \frac{f(b)-f(a)}{g(b)-g(a)}$ so that $\phi(a) = \phi(b)$. Apply Rolle's theorem to $\phi(x)$ to obtain $\phi'(c) = f'(c) - \lambda g'(c) = 0$.

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2. Taylor's Theorem with Remainder & Power Series

Taylor's Theorem with Remainder Terms:

If $f(x)$ and its first $(n-1)$ derivatives are continuous on $[a, a+h]$ and $f^{(n)}(x)$ exists on $(a, a+h)$, then:

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!}f''(a) + \cdots + \frac{h^{n-1}}{(n-1)!}f^{(n-1)}(a) + R_n$$

Where the remainder R_n can be expressed as:

- **Lagrange's Form of Remainder:** $\mathbf{R_n = \frac{h^n}{n!}f^{(n)}(a+\theta h)}, \quad (0 < \theta < 1)$
- **Cauchy's Form of Remainder:** $\mathbf{R_n = \frac{h^n}{(n-1)!}(1-\theta)^{n-1}f^{(n)}(a+\theta h)}, \quad (0 < \theta < 1)$

Maclaurin's Series Expansion (Centering at $a = 0$):

When $\lim_{n\to\infty} R_n = 0$, the function can be expressed as an infinite power series:

$$\mathbf{f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \cdots + \frac{x^n}{n!}f^{(n)}(0) + \ldots}$$

Standard Expansion Catalog for Exams (Expansion by Use of Known Series):

1. $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + \ldots \quad (\forall x \in \mathrm{R})$
2. $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots \quad (\forall x \in \mathrm{R})$
3. $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots \quad (\forall x \in \mathrm{R})$
4. $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots \quad (-1 < x \leq 1)$
5. $(1+x)^m = 1 + mx + \frac{m(m-1)}{2!}x^2 + \ldots \quad (|x| < 1)$

Problem:
Verify Rolle's Theorem for $f(x) = x^2 - 4x + 3$ on the interval $[1, 3]$.

Step 1 🚀: Check Rolle's Three Hypotheses

1. $f(x)$ is a polynomial, hence continuous on $[1, 3]$.

2. $f'(x) = 2x - 4$ exists on $(1, 3)$, hence differentiable on $(1, 3)$.

3. Check boundary values:

$$f(1) = 1^2 - 4(1) + 3 = 0$$

$$f(3) = 3^2 - 4(3) + 3 = 9 - 12 + 3 = 0 \implies f(1) = f(3)$$

Step 2 ⚡: Find Stationary Point $c \in (1, 3)$ where $f'(c) = 0$

$$f'(c) = 2c - 4 = 0 \implies \mathbf{c = 2}$$

Since $c = 2 \in (1, 3)$, Rolle's Theorem is **completely verified**!

🎯 Rolle's Theorem verified: $\mathbf{c = 2 \in (1, 3)}$.

University Question (4 Marks - MMDU Question Paper):
Expand $f(x) = e^x \cos x$ up to the term containing x^4 using Maclaurin's series (and verify by multiplication of known series).

Step 1 🚀: Compute Successive Derivatives at $x = 0$

- $f(x) = e^x \cos x \implies f(0) = 1$

- $f'(x) = e^x(\cos x - \sin x) \implies f'(0) = 1(1 - 0) = 1$

- $f''(x) = -2e^x \sin x \implies f''(0) = 0$

- $f'''(x) = -2e^x(\sin x + \cos x) \implies f'''(0) = -2(0 + 1) = -2$

- $f^{(4)}(x) = -4e^x \cos x \implies f^{(4)}(0) = -4(1) = -4$

Step 2 ⚡ : Substitute into Maclaurin's Formula (or Known Series)

$$e^x \cos x = (1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \dots)(1 - \frac{x^2}{2} + \frac{x^4}{24} - \dots) = 1 + x - \frac{x^3}{3} - \frac{x^4}{6} + \dots$$

🎯 Maclaurin Expansion: $\mathbf{e^x \cos x = 1 + x - \frac{x^3}{3} - \frac{x^4}{6} + O(x^5)}$.

2. Curvature, Centre & Chord of Curvature (Cartesian, Parametric, Polar)

1. Curvature, Radius & Center of Curvature: Comprehensive Theory

Physical Concept of Curvature (κ):

✅ **FUNDAMENTAL LAW / RULE:**
Definition: Let P and Q be two neighboring points on a smooth curve separated by arc length Δs. If the tangent vectors at P and Q make angles ψ and $\psi + \Delta\psi$ with the positive x-axis, the **Curvature** κ is the rate of turning of the tangent line per unit arc length:

$$\kappa = \lim_{\Delta s \to 0} \left| \frac{\Delta \psi}{\Delta s} \right| = \left| \frac{d\psi}{ds} \right|$$

Radius of Curvature (ρ):

The Radius of Curvature ρ is the reciprocal of the curvature:

$$\rho = \frac{1}{\kappa} = \left| \frac{ds}{d\psi} \right|$$

- A straight line has zero bending $\implies \kappa = 0 \implies \rho = \infty$.
- A circle of radius R bends uniformly $\implies \kappa = 1/R \implies \rho = R$.

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2. Analytical Formulas for Radius of Curvature ρ

Coordinate Representation	Curve Equation	Radius of Curvature Formula ρ
Cartesian $y = f(x)$	$y = f(x)$	$\rho = \frac{(1+y_1^2)^{3/2}}{y_2} \quad (y_1 = \frac{dy}{dx},\ y_2 = \frac{d^2y}{dx^2})$
Cartesian $x = g(y)$	$x = g(y)$	$\rho = \frac{(1+x_1^2)^{3/2}}{x_2} \quad (x_1 = \frac{dx}{dy},\ x_2 = \frac{d^2x}{dy^2})$
Parametric	$x = x(t),\ y = y(t)$	$\rho = \frac{(x'^2+y'^2)^{3/2}}{x'y''-y'x''}$
Polar Form	$r = f(\theta)$	$\rho = \frac{(r^2+r_1^2)^{3/2}}{r^2+2r_1^2-rr_2} \quad (r_1 = \frac{dr}{d\theta},\ r_2 = \frac{d^2r}{d\theta^2})$
Pedal Form (p-r)	$p = f(r)$	$\rho = r\frac{dr}{dp}$

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3. Center of Curvature $(\bar{x}, \bar{y})$ and Chord of Curvature

The circle that shares the same tangent and curvature as the curve at point $P(x, y)$ is the **Osculating Circle (Circle of Curvature)**. Its center $C(\bar{x}, \bar{y})$ is given by:

$$\bar{x} = x - \frac{y_1(1+y_1^2)}{y_2}, \quad \bar{y} = y + \frac{1+y_1^2}{y_2}$$

- **Equation of Circle of Curvature:** $(x-\bar{x})^2 + (y-\bar{y})^2 = \rho^2$.
- **Evolute:** The locus of the centers of curvature of all points on a curve is called the **Evolute** of that curve.

Chords of Curvature:

- **Chord parallel to x-axis:** $C_x = 2\rho\sin\psi = \frac{2y_1(1+y_1^2)}{y_2}$
- **Chord parallel to y-axis:** $C_y = 2\rho\cos\psi = \frac{2(1+y_1^2)}{y_2}$
- **Chord through the Pole (Polar Curve $r = f(\theta)$):** $C_p = 2\rho\sin\phi = \frac{2r(r^2+r_1^2)}{r^2+2r_1^2-rr_2}$

University Question (6 Marks):
Find the radius of curvature at any point t on the cycloid $x = a(t+\sin t),\ y = a(1-\cos t)$.

Step 1 🚀: Compute First and Second Derivatives w.r.t t

- $x' = \frac{dx}{dt} = a(1+\cos t) = 2a\cos^2(\frac{t}{2})$

- $y' = \frac{dy}{dt} = a\sin t = 2a\sin(\frac{t}{2})\cos(\frac{t}{2})$

- $x'' = -a\sin t, \quad y'' = a\cos t$

Step 2 ⚡ : Denominator $x'y'' - y'x''$ and Numerator $(x'^2 + y'^2)^{3/2}$

$$x'y'' - y'x'' = a^2(1+\cos t) = 2a^2\cos^2(\frac{t}{2})$$

$$(x'^2 + y'^2)^{3/2} = [4a^2\cos^2(\frac{t}{2})]^{3/2} = 8a^3\cos^3(\frac{t}{2})$$

$$\rho = \frac{8a^3\cos^3(t/2)}{2a^2\cos^2(t/2)} = 4a\cos(\frac{t}{2})$$

🎯 Radius of Curvature: $\rho = \mathbf{4a}\cos(\frac{\mathbf{t}}{\mathbf{2}})$.

University Question (6 Marks - MMDU Question Paper):
Find the coordinates of the centre of curvature $(\bar{X}, \bar{Y})$ and the equation of the evolute of the parabola $y^2 = 4ax$.

Step 1 🚀 : Derivatives for $y^2 = 4ax$

$$2yy_1 = 4a \implies y_1 = \frac{2a}{y}, \quad y_2 = -\frac{4a^2}{y^3}$$

Step 2 ⚡ : Centre of Curvature Formula

$$\bar{X} = x - \frac{y_1(1+y_1^2)}{y_2} = x + \frac{y^2 + 4a^2}{2a} = 3x + 2a$$

$$\bar{Y} = y + \frac{1+y_1^2}{y_2} = -\frac{y^3}{4a^2}$$

Step 3 🎯: Eliminate (x, y) to Obtain the Evolute Equation
Substitute $x = \frac{\bar{X}-2a}{3}$ and $y^2 = (4a^2\bar{Y})^{2/3}$ into $y^2 = 4ax$:

$$27a\bar{Y}^2 = 4(\bar{X} - 2a)^3$$

🎯 Centre of Curvature: $(\bar{X}, \bar{Y}) = (3x + 2a, -\frac{y^3}{4a^2})$ | Evolute: $\mathbf{27aY^2 = 4(X - 2a)^3}$.

3. Indeterminate Forms & L'Hospital's Rule

1. Rigorous Analysis of Indeterminate Forms & L'Hospital's Rule

What is an Indeterminate Form?

An algebraic limit expression $\lim_{x\to a} \frac{f(x)}{g(x)}$ is called **indeterminate** if substituting the limiting point $x = a$ fails to yield a well-defined value because the limiting behaviors of the numerator and denominator conflict.

The 7 Classic Indeterminate Forms:

$$\frac{\mathbf{0}}{\mathbf{0}}, \quad \frac{\infty}{\infty}, \quad \mathbf{0 \times \infty}, \quad \infty - \infty, \quad \mathbf{0^0}, \quad \mathbf{1^\infty}, \quad \infty^{\mathbf{0}}$$

2. L'Hospital's Rule Theorem & Derivation

✅ **FUNDAMENTAL LAW / RULE:**
Theorem (Bernoulli & L'Hospital): Let f and g be differentiable on an open interval containing a (except possibly at a). If $\lim_{x\to a} f(x) = 0$ and $\lim_{x\to a} g(x) = 0$ (or both tend to $\pm\infty$), and $g'(x) \neq 0$ near a, then:

$$\lim_{\mathbf{x\to a}} \frac{\mathbf{f(x)}}{\mathbf{g(x)}} = \lim_{\mathbf{x\to a}} \frac{\mathbf{f'(x)}}{\mathbf{g'(x)}}$$

3. Systematic Reduction Strategies for Non-Fractional Forms

1. **Form $0 \times \infty$:** Invert one factor into the denominator: $f(x) \cdot g(x) = \frac{f(x)}{1/g(x)} \implies \frac{0}{0}$.

2. **Form $\infty - \infty$:** Combine over a common denominator using LCM or rationalization.

3. **Exponential Forms ($0^0, 1^\infty, \infty^0$):** Let $y = [f(x)]^{g(x)}$, take $\ln y = g(x)\ln[f(x)]$, evaluate $L = \lim_{x\to a} \ln y$, and recover $\lim y = e^L$.

University Question (4 Marks):
Evaluate $\lim_{x\to 0} \left(\frac{1}{x^2} - \frac{1}{\sin^2 x}\right)$.

Step 1 🚀: Convert $\infty - \infty$ into $\frac{0}{0}$ via LCM

$$\lim_{x\to 0}\left(\frac{\sin^2 x - x^2}{x^2 \sin^2 x}\right) = \lim_{x\to 0} \frac{\sin^2 x - x^2}{x^4} \cdot \lim_{x\to 0}\left(\frac{x}{\sin x}\right)^2 = \lim_{x\to 0} \frac{\sin^2 x - x^2}{x^4}$$

Step 2 ⚡: Use Series Expansion $\sin x = x - \frac{x^3}{6} + O(x^5)$

$$\sin^2 x - x^2 = \left(x - \frac{x^3}{6}\right)^2 - x^2 = -\frac{x^4}{3} + O(x^6)$$

$$\lim_{x\to 0} \frac{-\frac{x^4}{3} + O(x^6)}{x^4} = -\frac{1}{3}$$

🎯 Evaluated Limit: $-\frac{1}{3}$.

University Question (6 Marks - MMDU Question Paper):
Evaluate $\lim_{x\to 0} \left(\frac{\tan x}{x}\right)^{1/x^2}$.

Step 1 🚀: Identify 1^∞ Form & Take Natural Logarithm
Let $y = \lim_{x\to 0} \left(\frac{\tan x}{x}\right)^{1/x^2} \implies \ln y = \lim_{x\to 0} \frac{\ln\left(\frac{\tan x}{x}\right)}{x^2}$.

Step 2 ⚡: Series Expansion of $\tan x$

$$\frac{\tan x}{x} = 1 + \frac{x^2}{3} + O(x^4) \implies \ln\left(\frac{\tan x}{x}\right) = \frac{x^2}{3} + O(x^4)$$

Step 3 🎯: Evaluate Limit and Exponentiate

$$\ln y = \lim_{x\to 0} \frac{\frac{x^2}{3} + O(x^4)}{x^2} = \frac{1}{3} \implies y = e^{1/3} = \sqrt[3]{e}$$

🎯 Limit: $\mathbf{e^{1/3} = \sqrt[3]{e}}$.

4. Beta and Gamma Functions & Their Properties

1. Rigorous Theory of Beta and Gamma Special Functions

A. The Gamma Function (Euler's Integral of Second Kind):

✅ **FUNDAMENTAL LAW / RULE:**
Definition: For any real number $n > 0$, the **Gamma Function** $\Gamma(n)$ is defined as:

$$\boldsymbol{\Gamma(n) = \int_0^\infty x^{n-1} e^{-x}\, dx \quad (n > 0)}$$

Fundamental Properties of $\Gamma(n)$:

1. **Recurrence Relation:** $\boldsymbol{\Gamma(n+1) = n\,\Gamma(n)}$ and $\boldsymbol{\Gamma(n+1) = n!}$ for positive integers.
2. **Base & Half-Integer Values:** $\Gamma(1) = 1$ and $\boldsymbol{\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}}$.

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2. The Beta Function (Euler's Integral of First Kind)

✅ **FUNDAMENTAL LAW / RULE:**
Definition: For $m > 0, n > 0$, the **Beta Function** $B(m, n)$ is defined as:

$$\boldsymbol{B(m, n) = \int_0^1 x^{m-1}(1-x)^{n-1}\, dx}$$

Key Properties and Master Identities:

1. **Symmetry:** $\mathbf{B(m, n) = B(n, m)}$.

2. **Trigonometric Form:**

$$\mathbf{B(m, n) = 2\int_0^{\pi/2} \sin^{2m-1}\theta \cos^{2n-1}\theta \, d\theta}$$

3. **Infinite Domain Form:**

$$\mathbf{B(m, n) = \int_0^{\infty} \frac{t^{m-1}}{(1+t)^{m+n}} \, dt}$$

4. **Beta-Gamma Bridge Relation:**

$$\mathbf{B(m, n) = \frac{\Gamma(m) \cdot \Gamma(n)}{\Gamma(m+n)}}$$

5. **Walli's / Trigonometric Integral Formula:**

$$\int_0^{\pi/2} \sin^{p}\theta \cos^{q}\theta \, d\theta = \frac{\Gamma\left(\frac{p+1}{2}\right)\Gamma\left(\frac{q+1}{2}\right)}{2\,\Gamma\left(\frac{p+q+2}{2}\right)}$$

6. **Legendre's Duplication Formula:**

$$\Gamma(m)\,\Gamma\left(m+\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2^{2m-1}}\,\Gamma(2m)$$

University Question (6 Marks - MMDU Question Paper):
Evaluate $\int_0^{\pi/2} \sin^4\theta \cos^6\theta \, d\theta$ using Beta and Gamma functions, and prove $\int_0^1 \frac{dx}{\sqrt{1-x^4}} = \frac{\sqrt{\pi}\,\Gamma(1/4)}{4\,\Gamma(3/4)}$.

Step 1 🚀: Evaluate Trigonometric Integral ($p = 4, q = 6$)

$$I_1 = \frac{\Gamma(\frac{5}{2})\Gamma(\frac{7}{2})}{2\,\Gamma(6)} = \frac{(\frac{3\sqrt{\pi}}{4})(\frac{15\sqrt{\pi}}{8})}{2 \times 120} = \frac{3\pi}{512}$$

Step 2 ⚡: Prove $\int_0^1 \frac{dx}{\sqrt{1-x^4}} = \frac{\sqrt{\pi}\,\Gamma(1/4)}{4\,\Gamma(3/4)}$
Substitute $x^4 = u \implies x = u^{1/4},\ dx = \frac{1}{4}u^{-3/4}du$:

$$I_2 = \frac{1}{4}\int_0^1 u^{1/4-1}(1-u)^{1/2-1}du = \frac{1}{4}B(\frac{1}{4},\frac{1}{2}) = \frac{\Gamma(1/4)\Gamma(1/2)}{4\,\Gamma(3/4)} = \frac{\sqrt{\pi}\,\Gamma(1/4)}{4\,\Gamma(3/4)}$$

🎯 Integral Value: $\frac{3\pi}{512}$ | Formula proved via $B(\frac{1}{4},\frac{1}{2})$.

University Solved Examination Questions (Unit 2)

MMDU Official University Solved Examination Questions (Unit 2)

Course: Mathematics I (`BMAT-001`)
Pattern: Sections A (1 Mark Compulsory), B (2 Marks Analytical), C (4 Marks Short Long), D (6 Marks Long)

📘 Question 4 (1 Mark): Geometric Interpretation of Rolle's Theorem

Question:
State the geometric meaning of the point c in Rolle's Theorem.

Step-by-Step Solution:
If a continuous, differentiable curve $y = f(x)$ has equal values at the endpoints $f(a) = f(b)$, there exists at least one interior point $c \in (a, b)$ where $f'(c) = 0$, meaning the **tangent line at** $(c, f(c))$ **is parallel to the** x**-axis**.

🎯 **Verified Result:** Horizontal tangent parallel to x-axis at $x = c$.

📘 Question 14 (2 Marks): Radius of Curvature of Catenary Curve

Question:

Find the radius of curvature ρ for the catenary $y = c\cosh(x/c)$.

Step-by-Step Solution:

Given $y = c\cosh(x/c)$, we have $y_1 = \sinh(x/c)$ and $y_2 = \frac{1}{c}\cosh(x/c)$.
Since $1 + y_1^2 = 1 + \sinh^2(x/c) = \cosh^2(x/c)$:

$$\rho = \frac{(1+y_1^2)^{3/2}}{y_2} = \frac{\cosh^3(x/c)}{\frac{1}{c}\cosh(x/c)} = c\cosh^2(x/c) = \frac{y^2}{c}$$

🎯 **Verified Result:** $\rho = \frac{y^2}{c}$.

📘 Question 22 (6 Marks): Relation Between Beta and Gamma Functions

Question:

State and prove $B(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$ and hence evaluate $\int_0^{\pi/2} \sqrt{\tan\theta}\, d\theta$.

Step-by-Step Solution:

Using polar coordinates $x = r\cos\theta, y = r\sin\theta$ in $\Gamma(m)\Gamma(n) = 4\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} x^{2m-1} y^{2n-1}\, dx\, dy$ yields $\Gamma(m)\Gamma(n) = B(m,n)\Gamma(m+n)$.
For $\int_0^{\pi/2} \sin^{1/2}\theta \cos^{-1/2}\theta\, d\theta$: here $p = 1/2, q = -1/2$, so:

$$I = \frac{\Gamma(3/4)\Gamma(1/4)}{2\Gamma(1)} = \frac{1}{2}\cdot\frac{\pi}{\sin(\pi/4)} = \frac{\pi}{\sqrt{2}}$$

🎯 **Verified Result:** $\int_0^{\pi/2} \sqrt{\tan\theta}\, d\theta = \frac{\pi}{\sqrt{2}}$.